

How would you boot into Linux using your USB thumb drive?

Send in your solution with the subject "Take a Crack", and your postal address, to takeacrack@thinkdigit.com

Get Tux onto your USB drive



LAST MONTH'S CHALLENGE

Win!



Take a Crack and win

Good Programming

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LAST MONTH'S WINNER

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Rules and Regulations

Readers are requested to send in their answers by the 15th of the month of publication.

Employees of Jasubhai Digital Media and their relatives are not permitted to participate in this contest.

Readers are encouraged to send their replies by e-mail. Jasubhai Digital Media will not entertain any unsolicited communication.

Jasubhai Digital Media is not responsible for any damage to your system that may be caused while you are trying to solve the problem.

Bypass Windows File Protection in Windows XP

There are two different solutions, depending on whether you have SP2 installed or not.

If you don't have SP2 installed

Locate the file `sfc_os.dll`, which should be in your System32 folder. Copy it and name it `sfc_os.bak`. Open this file using a hex editor such as XVI32, which is an excellent freeware. This is an extremely simple edit and does not require an editor with advanced features. All you need to do is change two values.

The location of these values will vary depending on whether you're using Windows XP without a Service Pack or Windows XP with SP1, but the actual hex values to change will be the same.

For Windows XP with no Service Pack installed, go to offset 0000E2B8 (E2B8 hex). For Windows XP with SP1, go to offset 0000E3BB (E3BB hex). At those offsets, change the hex values 8BC6 to 9090.

Note: Depending on your hex editor, you may see the value separated by a space, as in "8B C6".

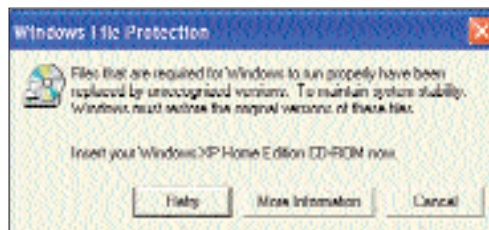
If you can't find the values, do not proceed. It could be that your version of XP is different. The edit as described here only applies to English retail versions of Windows XP.



Set the Registry key to disable SFC

If you have SP2 installed

Disabling System File Protection in XP prior to SP2 involved editing the `sfc_os.dll` file, but the version of `sfc_os.dll` that comes with SP2 is different, and you cannot disable file protection with it. With SP2, you need to use the previous version of the `sfc_os.dll` file, which is 5.1.2600.1106 (the one with SP2 is 5.1.2600.2180).



The Windows File Protection error

Open `sfc_os.dll` with a hex editor and go to offset 0xECE9. The values to change are 33 C0 40. Change these to 90 90 90. Save your changes. Rename the file with a .bak extension (`sfc_os.bak`).

The original exists in two locations: the `\Windows\System32` folder, and the `\Windows\System32\dlcache` folder. Place a copy of `sfc_os.bak` in both these folders. Then in the `dlcache` folder, look for `cmd.exe`, double-click it, and enter this command:

```
copy SFC_OS.BAK SFC_OS.DLL /Y
```

Next, go to your System32 folder and do the same thing. Look for `cmd.exe`, double-click it, and enter exactly the same command.

Points to remember

- ❑ If you have a problem overwriting the files, you can do the copying within the Recovery Console, or you can try Safe Mode.
- ❑ The `dlcache` folder is a hidden folder.
- ❑ It is important to copy the file to your `dlcache` folder *first*.
- ❑ If you're prompted to pop in your Windows CD, click Cancel.

Once the files have been copied, shut down your computer and restart it. The final step is to disable System File Protection in the Registry. Go to `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon`. In the right-hand pane, if it doesn't exist, create a new DWORD value named `SFCDisable`. Double-click it and enter "FFFFFF9D". Exit the Registry Editor and reboot. You're done. ■